

Lecture 14:

Classification, Statistical

Sins

Announcements

- Reading
 - Chapter 21
- Course evaluations
 - Online evaluation now through noon on Friday, December 16
- Will be making study code for final exam available later today

Compare to KNN Results (from Monday)

Average of 10 80/20 splits using KNN (k=3)

Accuracy = 0.744

Sensitivity = 0.629

Specificity = 0.829

Pos. Pred. Val. = 0.728

Average of LOO testing using KNN (k=3)

Accuracy = 0.769

Sensitivity = 0.663

Specificity = 0.842

Pos. Pred. Val. = 0.743

Average of 10 80/20 splits LR

Accuracy = 0.804

Sensitivity = 0.719

Specificity = 0.859

Pos. Pred. Val. = 0.767

Average of LOO testing using LR

Accuracy = 0.786

Sensitivity = 0.705

Specificity = 0.842

Pos. Pred. Val. = 0.754

Performance not much difference

Logistic regression slightly better

Logistic regression provides insight about variables

Looking at Feature Weights

`model.classes_ = ['Died' 'Survived']`

For label Survived

`C1 = 1.66761946545`

`C2 = 0.460354552452`

`C3 = -0.50338282535`

`age = -0.0314481062387`

`male gender = -2.39514860929`

Be wary of reading too
much into the weights

Features are often
correlated

L1 regression tends to drive one variable to zero

L2 (default) regression spreads weights across variables

Correlated Features, an Example

- $c1 + c2 + c3 = 1$
 - I.e., values are not independent
 - Is being in 1st class good, or being in the other classes bad?
- Suppose we eliminate $c1$?

```
def __init__(self, pClass, age, gender, survived, name):  
    self.name = name  
    if pClass == 2:  
        self.featureVec = [1, 0, age, gender]  
    elif pClass == 3:  
        self.featureVec = [0, 1, age, gender]  
    else:  
        self.featureVec = [0, 0, age, gender]  
    self.label = survived  
    self.cabinClass = pClass
```

Comparative Results

Original Features

Average of 20 80/20 splits LR

Accuracy = 0.778

Sensitivity = 0.687

Specificity = 0.842

Pos. Pred. Val. = 0.755

model.classes_ = ['Died' 'Survived']

For label Survived

C1 = 1.68864047459

C2 = 0.390605976351

C3 = -0.46270349333

age = -0.0307090135358

male gender = -2.41191131088

Modified Features

Average of 20 80/20 splits LR

Accuracy = 0.779

Sensitivity = 0.674

Specificity = 0.853

Pos. Pred. Val. = 0.765

model.classes_ = ['Died' 'Survived']

For label Survived

C2 = -1.08356816806

C3 = -1.92251427055

age = -0.026056041377

male gender = -2.36239279331

Changing the Cutoff

```
random.seed(0)
trainingSet, testSet = split80_20(examples)
model = buildModel(trainingSet, False)
print('Try p = 0.1')
truePos, falsePos, trueNeg, falseNeg = \
    applyModel(model, testSet, 'Survived', 0.1)
getStats(truePos, falsePos, trueNeg, falseNeg)
print('Try p = 0.9')
truePos, falsePos, trueNeg, falseNeg = \
    applyModel(model, testSet, 'Survived', 0.9)
getStats(truePos, falsePos, trueNeg, falseNeg)
```

Try p = 0.1

Accuracy = 0.493

Sensitivity = 0.976

Specificity = 0.161

Pos. Pred. Val. = 0.444

Try p = 0.9

Accuracy = 0.656

Sensitivity = 0.176

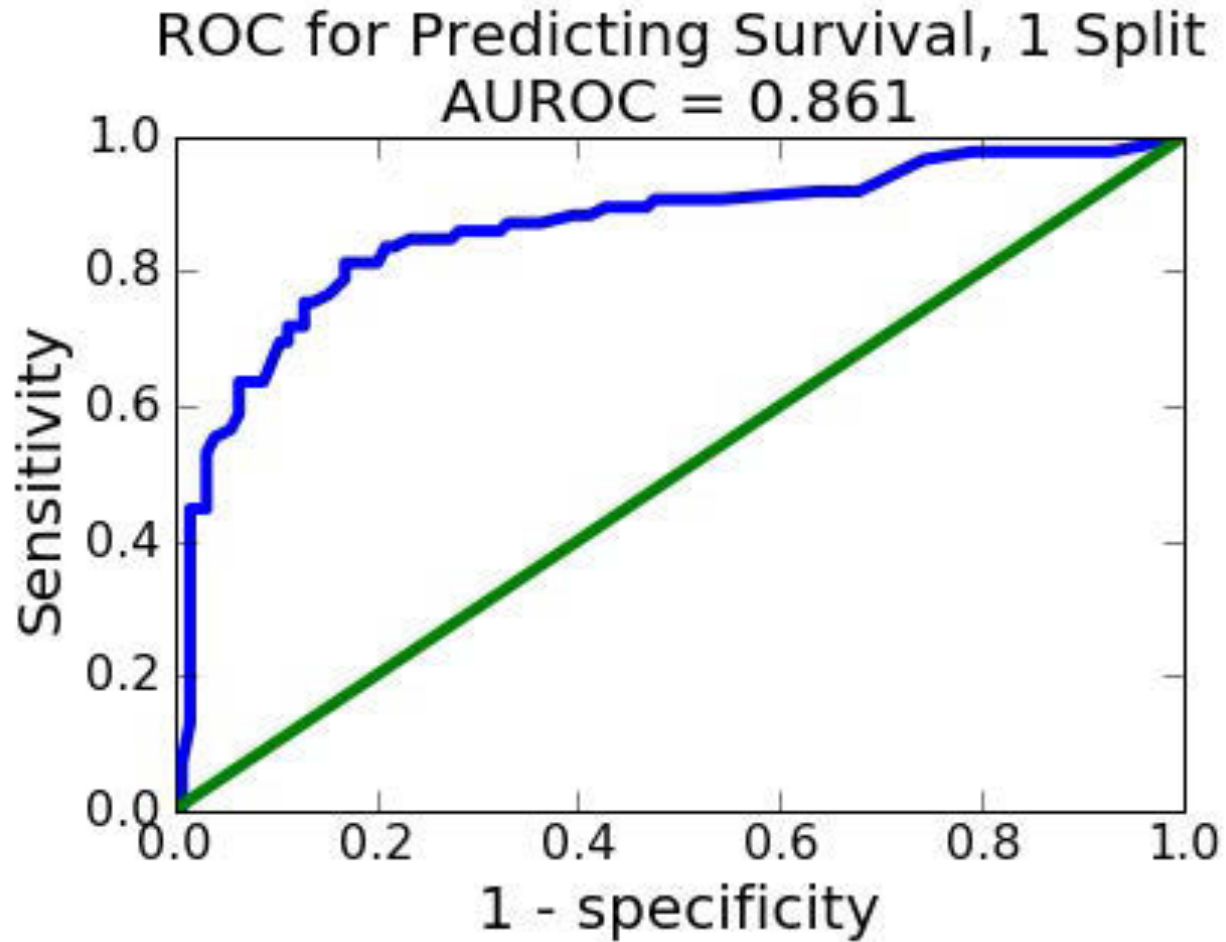
Specificity = 0.984

Pos. Pred. Val. = 0.882

ROC (Receiver Operating Characteristic)

```
def buildROC(trainingSet, testSet, title, plot = True):
    model = buildModel(trainingSet, True)
    xVals, yVals = [], []
    p = 0.0
    while p <= 1.0:
        truePos, falsePos, trueNeg, falseNeg = \
            applyModel(model, testSet,
                       'Survived', p)
        xVals.append(1.0 - specificity(trueNeg, falsePos))
        yVals.append(sensitivity(truePos, falseNeg))
        p += 0.01
    auroc = sklearn.metrics.auc(xVals, yVals, True)
    if plot:
        ...|
    return auroc
```


Output



There are Three Kinds of Lies

LIES

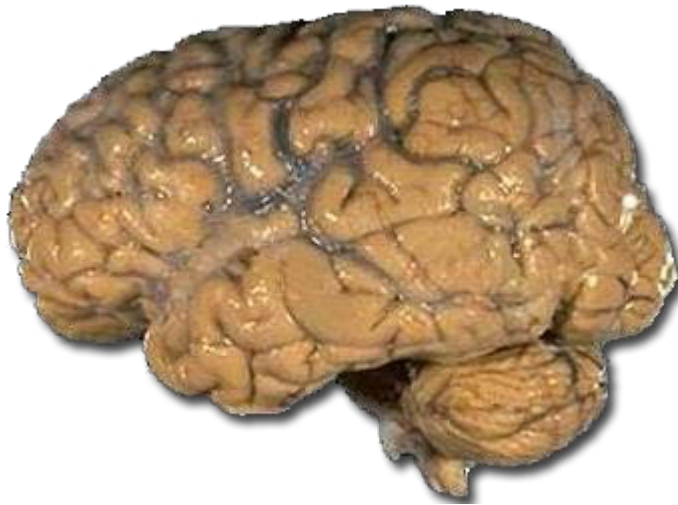
DAMNED LIES

and

STATISTICS

Humans and Statistics

Human Mind



Statistics

$$\sum_{i=0}^{i=\text{len}(\text{observed})-1} (\text{observed}[i] - \text{predicted}[i])^2$$
$$1 - \left(\frac{n-1}{n} * \frac{n-2}{n} * \dots * \frac{n-(K-1)}{n} \right)$$

Image of brain © source unknown. All rights reserved. This content is excluded from our Creative Commons license. For more information, see <https://ocw.mit.edu/help/faq-fair-use/>.

Humans and Statistics

“If you can't prove what you want to prove, demonstrate something else and pretend they are the same thing. In the daze that follows the collision of statistics with the human mind, hardly anyone will notice the difference.” – *Darrell Huff*

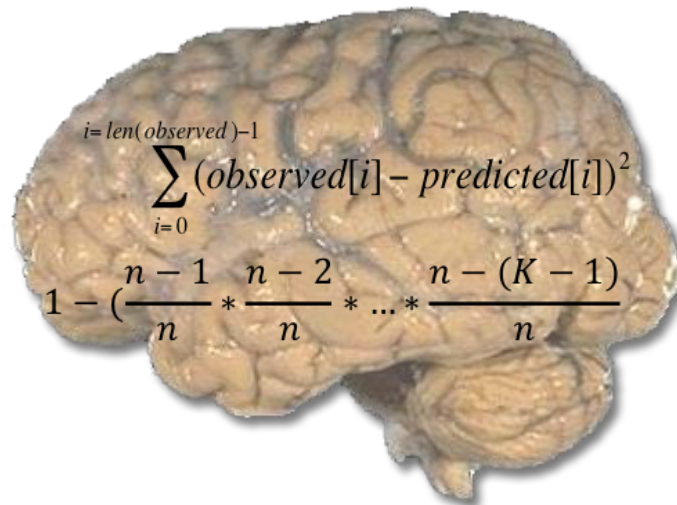


Image of brain © source unknown. All rights reserved. This content is excluded from our Creative Commons license. For more information, see <https://ocw.mit.edu/help/faq-fair-use/>.

Anscombe's Quartet

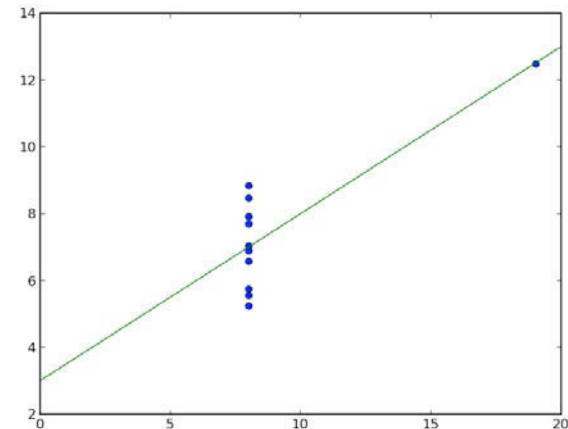
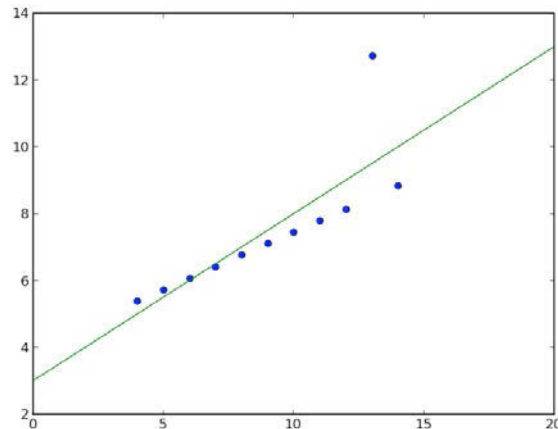
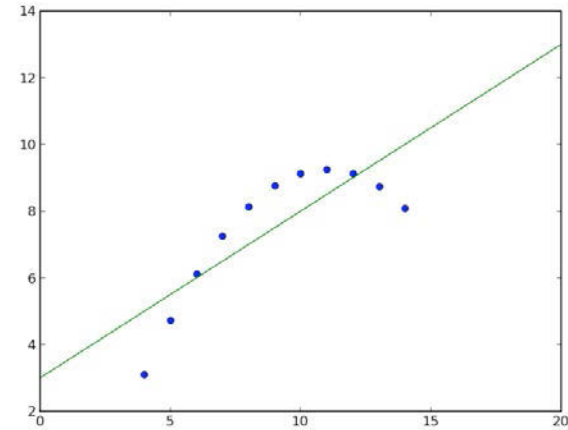
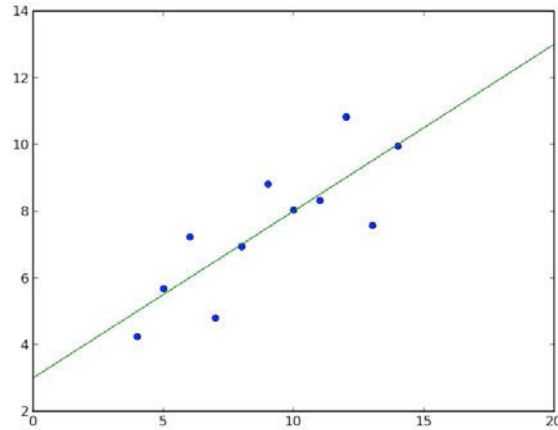
- Four groups each containing 11 x, y pairs

x	y	x	y	x	y	x	y
10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
5.0	5.68	5.0	4.74	5.0	5.73	8.0	6.89

Summary Statistics

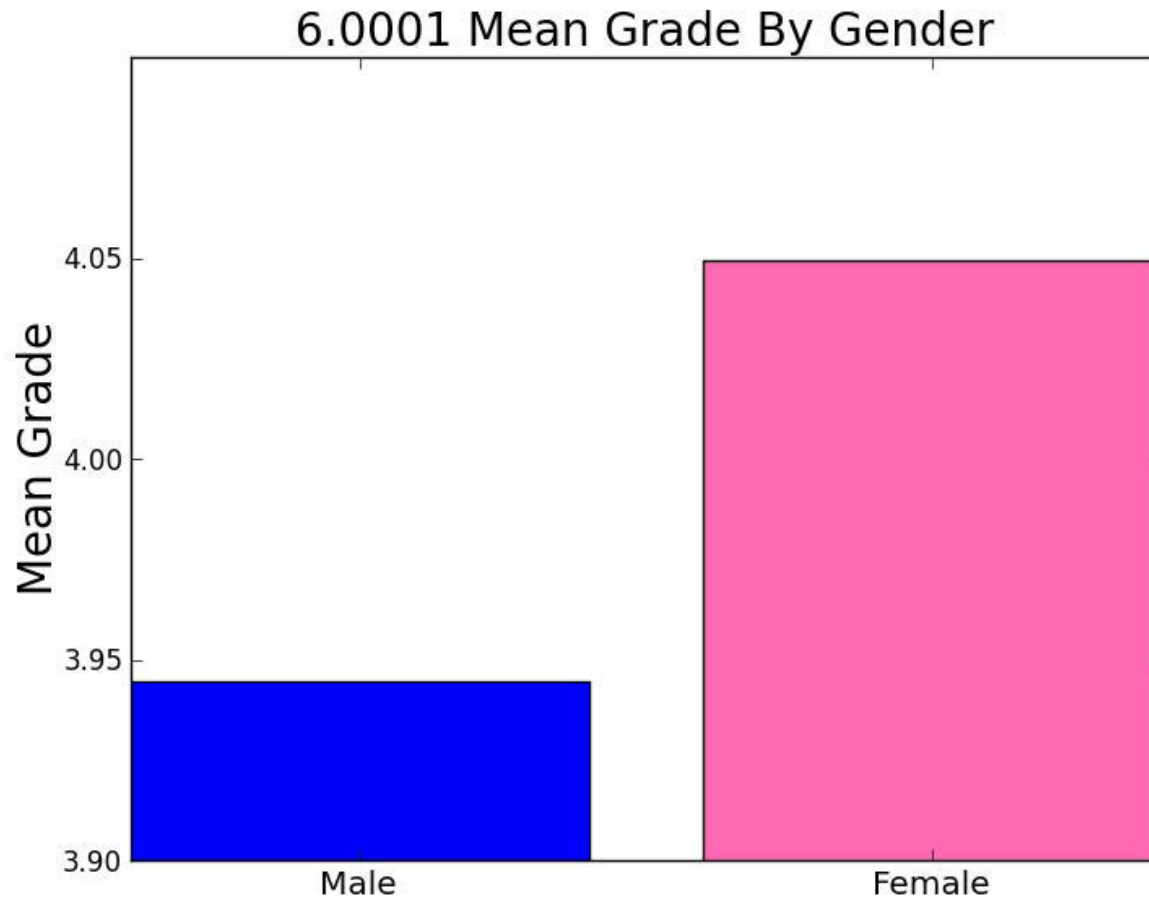
- Summary statistics for groups identical
 - Mean $x = 9.0$
 - Mean $y = 7.5$
 - Variance of $x = 10.0$
 - Variance of $y = 3.75$
 - Linear regression model: $y = 0.5x + 3$
- Are four data sets really similar?

Let's Plot the Data

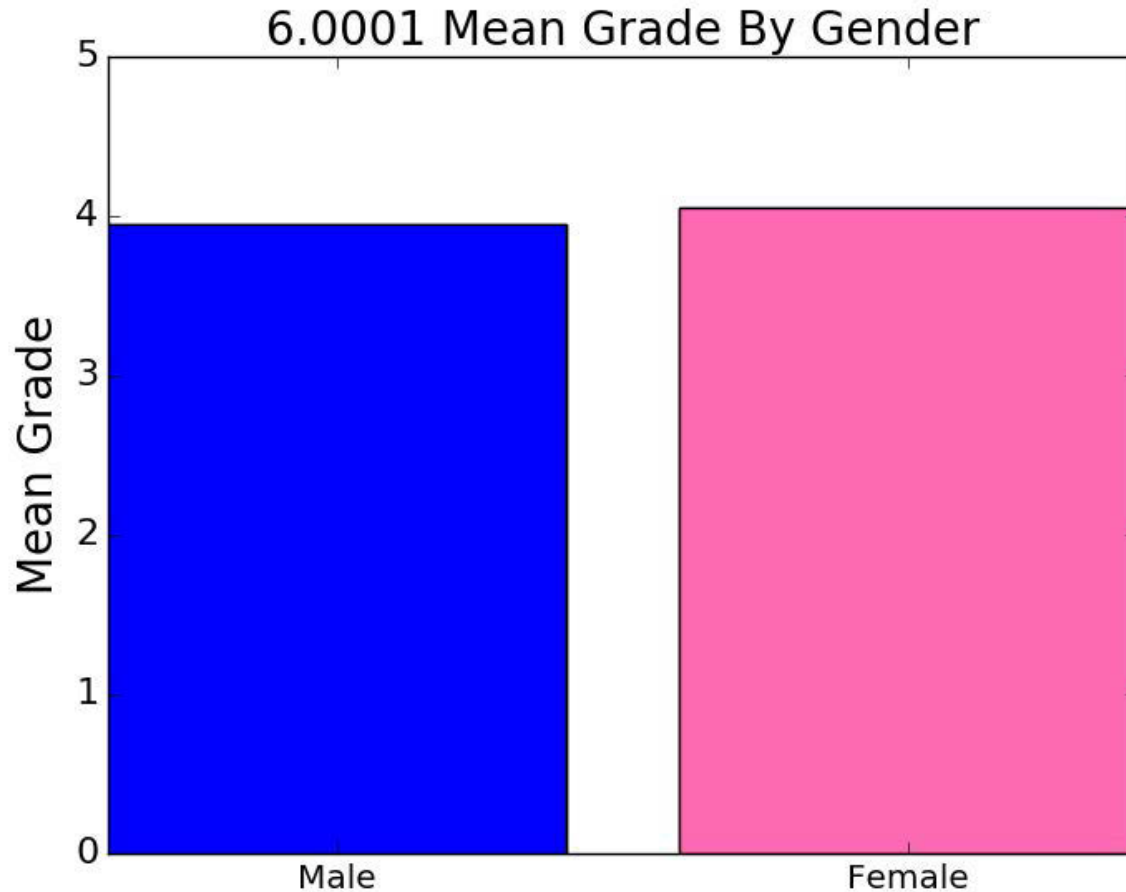


Moral: Statistics about the data is not the same as the data
Moral: Use visualization tools to look at the data itself

Lying with Pictures



Telling the Truth with Pictures



Moral: Look carefully at the axes labels and scales

Lying with Pictures



Moral: Ask whether the things being compared are actually comparable

Screenshot of Fox News © 20th / 21st Century Fox. All rights reserved. This content is excluded from our Creative Commons license. For more information, see <https://ocw.mit.edu/help/faq-fair-use/>.

Garbage In, Garbage Out

“On two occasions I have been asked [by members of Parliament], ‘Pray, Mr. Babbage, if you put into the machine wrong figures, will the right answers come out?’ I am not able rightly to apprehend the kind of confusion of ideas that could provoke such a question.” – Charles Babbage (1791-1871)

Calhoun's Response to Errors in Data

“there were so many errors they balanced one another, and led to the same conclusion as if they were all correct.”

Was it the case that the measurement errors are unbiased and independent of each of other, and therefore almost identically distributed on either side of the mean?

No, later analysis showed that the errors were not random but systematic.

“it was the census that was insane and not the colored people.” — James Freeman Clarke

Moral: Analysis of bad data can lead to dangerous conclusions.

Sampling

- All statistical techniques are based upon the assumption that by sampling a subset of a population we can infer things about the population as a whole
- As we have seen, *if random sampling is used*, one can make meaningful mathematical statements about the expected relation of the sample to the entire population
- Easy to get random samples in simulations
- Not so easy in the field, where some examples are more convenient to acquire than others

Non-representative Sampling

- “Convenience sampling” not usually random, e.g.,
 - Survivor bias, e.g., course evaluations at end of course or grading final exam in 6.0002 on a strict curve
 - Non-response bias, e.g., opinion polls conducted by mail or online
- When samples not random and independent, we can still do things like computer means and standard deviations, but **we should not draw conclusions from them** using things like the empirical rule and central limit theorem.
- **Moral: Understand how data was collected, and whether assumptions used in the analysis are satisfied. If not, be wary.**

MIT OpenCourseWare

<https://ocw.mit.edu>

6.0002 Introduction to Computational Thinking and Data Science

Fall 2016

For information about citing these materials or our Terms of Use, visit: <https://ocw.mit.edu/terms>.